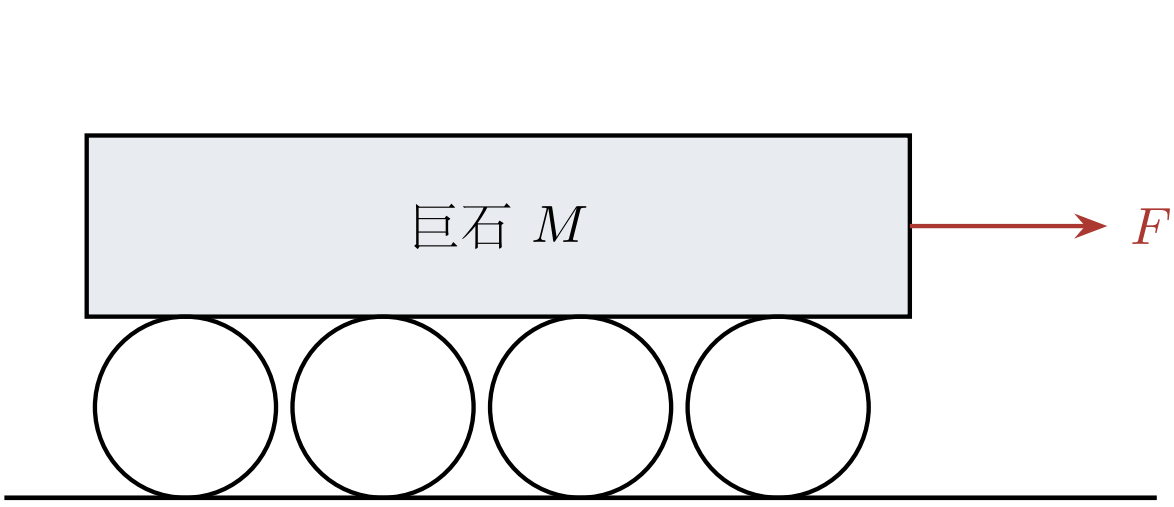
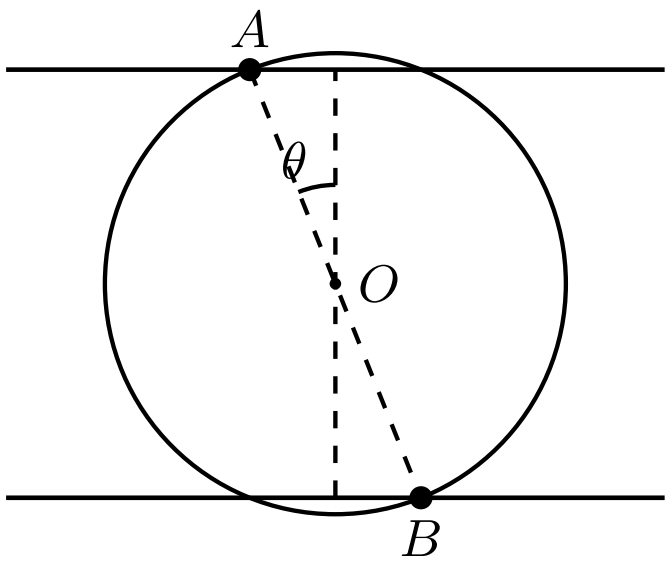
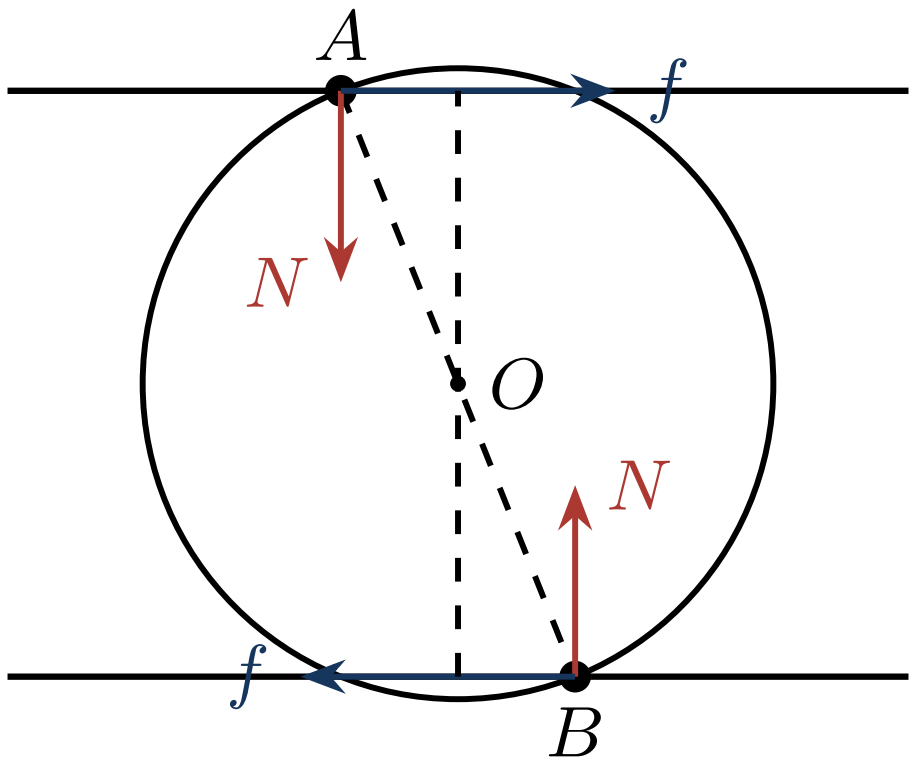


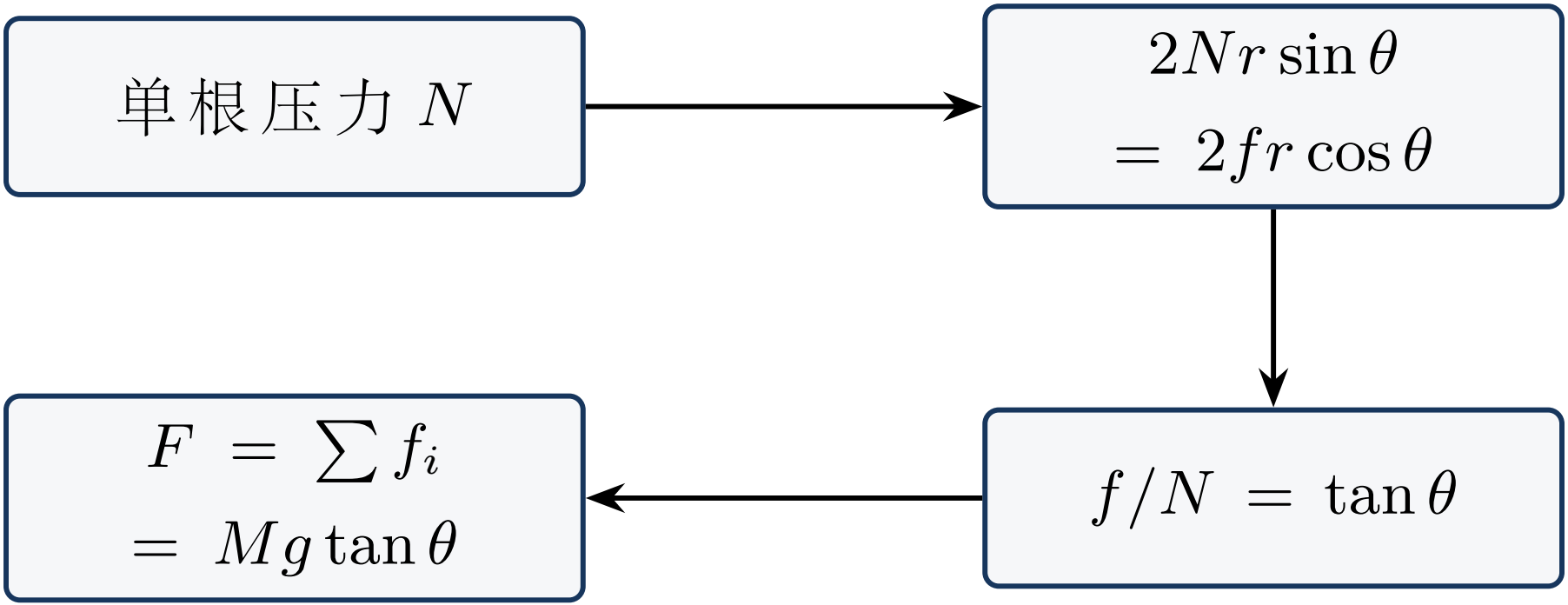

10 根圆木（数量示意）



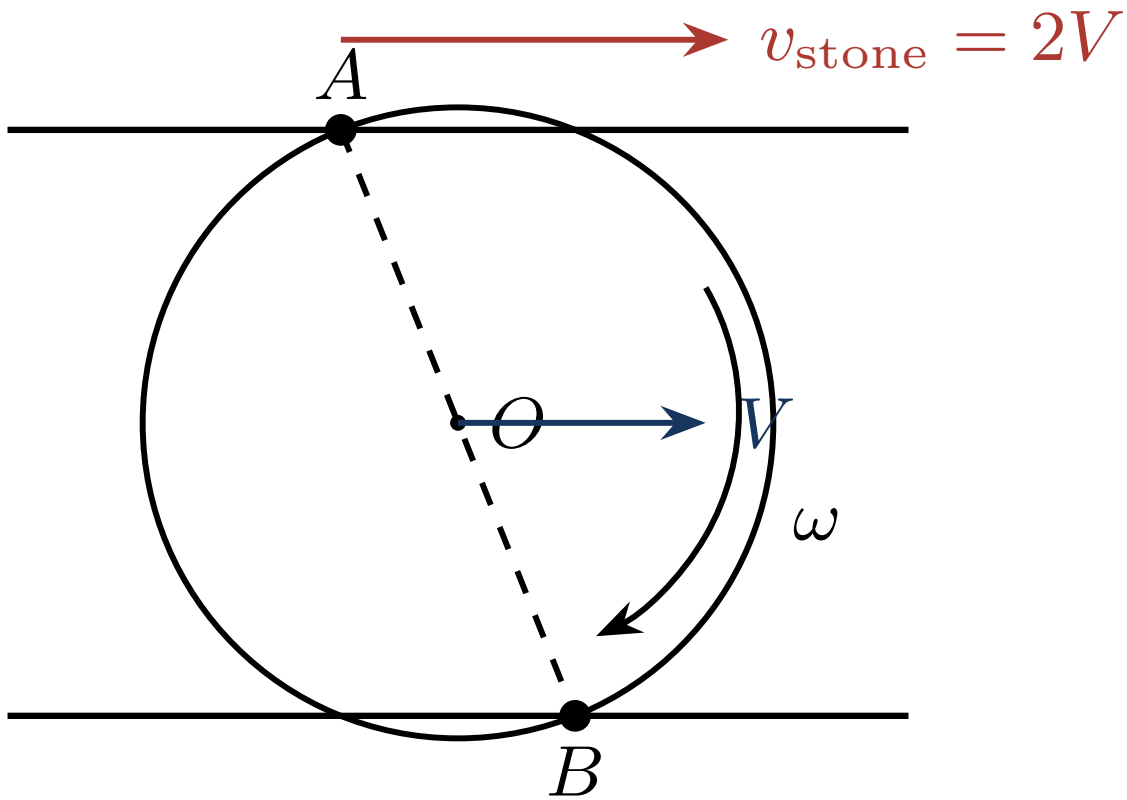

压缩后竖直间距为 $2r(1-\eta)$。图中偏移角夸大以便辨认。



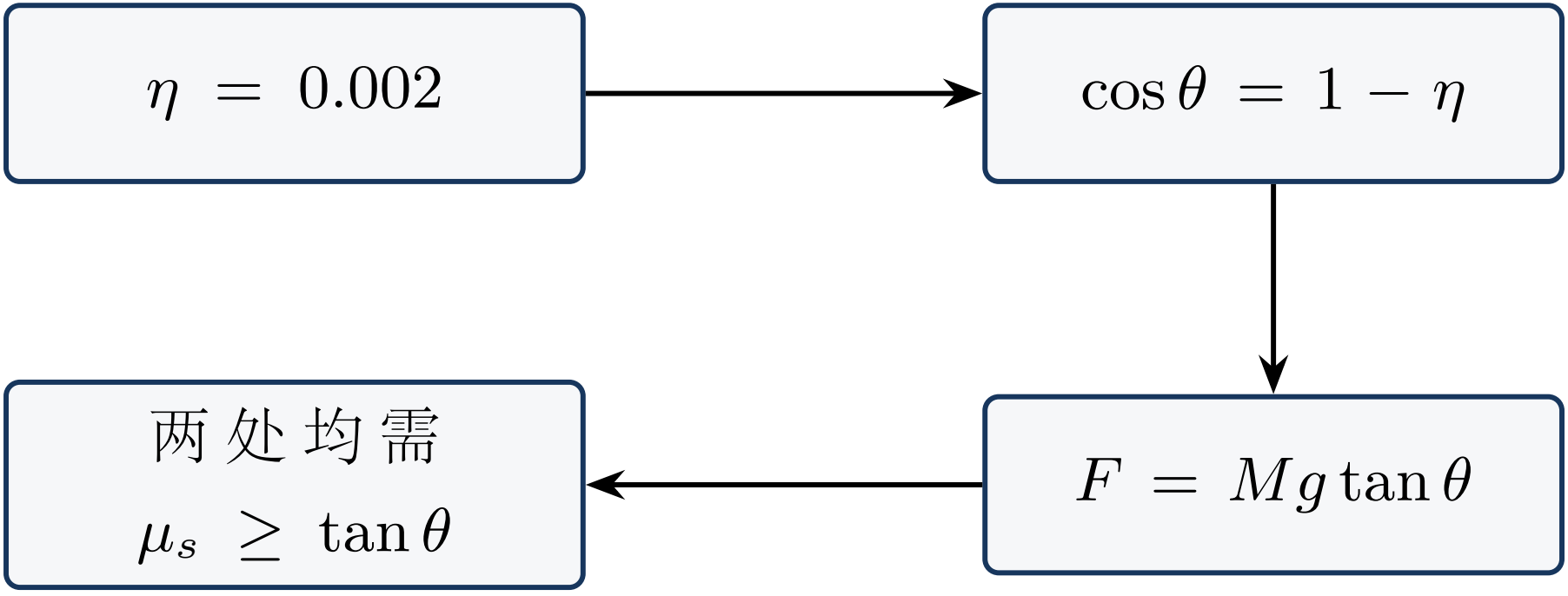

压力偏移力矩 $2Nr\sin\theta$；切向力矩 $2fr\cos\theta$，两者相反。




压力与切向力各自合力为零，但两对力矩要相互抵消。




下接触：$V - \omega r\cos\theta = 0$；上接触：$V + \omega r\cos\theta = 2V$。




结果依赖题设压力合力偏移模型。无滑动不等于切向力为零；形状恢复不等于能量无损。